

GEMFIBROZIL (LOPID)

Pharma Talk



Fibric Acid Derivative | PPAR-α Agonist

Lowers triglycerides and raises HDL by activating lipid metabolism.

1 MECHANISM OF ACTION (MOA) & TARGET



- **TARGET:** PPAR-α
- Activates receptors
  → ↑ lipoprotein lipase
  → ↓ TG & ↑ HDL
- **Analogy:** Turns on fat-burning enzymes to clear lipids.



2 FORMS & STRENGTHS



- **Tablet:** 600 mg

3 INDICATIONS (FDA-APPROVED & MAJOR OFF-LABEL)

FDA-APPROVED (ADULTS)

- Severe hypertriglyceridemia (Types IV, V) at high risk for pancreatitis unresponsive to diet. Diet adjunct.
- Primary dysbetalipoproteinemia (Type III) unresponsive to diet. Diet adjunct.

MAJOR OFF-LABEL USES

- Mixed dyslipidemia in patients intolerant to statins.
- Cardiovascular risk reduction in specific phenotypes.

4 DOSAGE & ADMINISTRATION

A) ADULT DOSE (ORAL)

- **Standard:** 600 mg twice daily
- **Timing:** 30 min before breakfast & dinner
- **Max dose:** 1200 mg/day

B) PEDIATRIC DOSE

- **PEDIATRIC USE** — Safety/efficacy not established.

C) ADMINISTRATION

- **Route:** Orally
- Take exactly 30 min before breakfast & dinner.
- Swallow whole with water.

D) ADJUSTMENTS

- **Renal:** Avoid in severe impairment.
- **Hepatic:** Contraindicated in primary biliary cirrhosis.

5 PHARMACOKINETICS & DYNAMICS (PK/PD)

ONSET	HALF-LIFE	METABOLISM	ELIMINATION
2–5 days	1.5 hours	Hepatic glucuronidation	Renal ~70%

MOA RESULT: ↓ TG, ↑ HDL, slight ↓ LDL.

6 ADVERSE EFFECTS & TOXICITY

COMMON

- Dyspepsia
- Abdominal pain
- Diarrhea
- Fatigue
- Nausea



SEVERE (STOP & REPORT)

- Rhabdomyolysis
- Cholelithiasis
- Hepatotoxicity
- Myositis



7 CONTRAINDICATIONS & INTERACTIONS

ABSOLUTE CONTRAINDICATIONS

- Hepatic disease
- Severe renal dysfunction
- Gallbladder disease

BOXED WARNING

NONE

MAJOR DRUG INTERACTIONS

- Simvastatin → ↓ glucuronidation → Rhabdomyolysis
- Warfarin → Displacement from protein binding → ↑ bleeding risk
- Repaglinide → CYP2C8 inhibition → Severe hypoglycemia

8 SPECIAL POPULATIONS (PREGNANCY & LACTATION)

- **Pregnancy:** Category C
- **Lactation:** Avoid
- **Geriatric:** Use lower dose; monitor renal function closely.

9 CLINICAL PEARLS



1. Never combine with statins (especially simvastatin) due to severe myopathy risk.
2. Take exactly 30 min before meals for optimal absorption.
3. Monitor LFTs baseline & periodically; check CK if myalgia occurs.
4. Discontinue if gallstones develop.
5. Assess renal function before initiation.
6. Fibrates less effective than statins for LDL reduction but superior for TG lowering.
7. Counsel patients to report unexplained muscle pain.

10 GLOSSARY OF ABBREVIATIONS

- **TG** — Triglycerides
- **HDL** — High-Density Lipoprotein
- **LFTs** — Liver Function Tests
- **CK** — Creatine Kinase
- **CYP** — Cytochrome P450
- **LDL** — Low-Density Lipoprotein
- **PPAR-α** — Peroxisome Proliferator-Activated Receptor alpha